

Math 520, F07
Quiz 9

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (20 points) Solve the following initial-value problem:

$$\frac{dx}{dt} = \begin{pmatrix} -1 & -4 & 2 \\ 0 & -1 & 3 \\ 0 & 2 & -2 \end{pmatrix} x, \quad x(0) = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Solution: $A = \begin{pmatrix} -1 & -4 & 2 \\ 0 & -1 & 3 \\ 0 & 2 & -2 \end{pmatrix}$

The characteristic polynomial is

$$P(\lambda) = \det(A - \lambda I) \\ = \det \begin{pmatrix} -1-\lambda & -4 & 2 \\ 0 & -1-\lambda & 3 \\ 0 & 2 & -2-\lambda \end{pmatrix}$$

$$= (-1-\lambda)[(-1-\lambda)(-2-\lambda)-6]$$

$$= -(\lambda+1)(\lambda^2+3\lambda-4)$$

$$= -(\lambda+1)(\lambda-1)(\lambda+4) = 0$$

$$\Rightarrow \lambda_1 = -1, \lambda_2 = 1, \lambda_3 = -4.$$

i) $\lambda_1 = -1:$

$$(A + I)V = 0 \Rightarrow \begin{bmatrix} 0 & -4 & 2 \\ 0 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} -4V_2 + 2V_3 = 0 \\ 3V_3 = 0 \end{cases} \Rightarrow V_2 = V_3 = 0$$

$$2V_2 - V_3 = 0$$

$$V = \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} V_1 \\ 0 \\ 0 \end{pmatrix} = V_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = C \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow x_1(t) = e^{\lambda_1 t} V = e^{-t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

ii) $\lambda_2 = 1:$

$$(A - I)V = 0 \Rightarrow \begin{bmatrix} -2 & -4 & 2 \\ -2 & 3 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} -2V_1 - 4V_2 + 2V_3 = 0 \\ -2V_1 + 3V_3 = 0 \end{cases} \Rightarrow V_1 = -2V_3$$

$$V = \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} -2V_3 \\ \frac{3}{2}V_3 \\ V_3 \end{pmatrix} = V_3 \begin{pmatrix} -2 \\ \frac{3}{2} \\ 1 \end{pmatrix} = C \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix}$$

$$\Rightarrow x_2(t) = e^{\lambda_2 t} V = e^t \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix}$$

iii) $\lambda_3 = -4:$

$$(A + 4I)V = 0 \Rightarrow \begin{bmatrix} 3 & -4 & 2 \\ 0 & 3 & 3 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} 3V_1 - 4V_2 + 2V_3 = 0 \\ 3V_2 + 3V_3 = 0 \end{cases} \Rightarrow V_1 = -2V_3$$

$$V = \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} -2V_3 \\ -V_3 \\ V_3 \end{pmatrix} = V_3 \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = C \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow x_3(t) = e^{-4t} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

Thus, the general solution is

$$x(t) = C_1 e^{-t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + C_2 e^t \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix} + C_3 e^{-4t} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

From the initial values.

$$x(0) = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + C_2 \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix} + C_3 \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{cases} C_1 - 4C_2 - 2C_3 = 0 \\ 3C_2 - C_3 = -1 \\ 2C_3 + C_3 = 1 \end{cases} \Rightarrow \begin{cases} C_1 = 2 \\ C_2 = 0 \\ C_3 = 1 \end{cases}$$

$$\text{Thus, } x(t) = 2e^{-t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + e^{-4t} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$